

Power Quality Enhancement in Integrated Grid for EVs Charging Stations Using Active Filter

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Abstract— The concern over greenhouse gas emissions and their role in global warming is one of the main drivers behind the introduction of electric cars into the market. There are several reasons for users to prefer electric cars (EVs), such as improved performance, reduced environmental impact from zero carbon emissions, and other benefits. Energy sustainability requires consumers to be aware of the environment and to have a vision for renewable energy. This article is related to the designing of charging station with active filter for mitigation of total harmonic distortion. The purpose of this article is to reduce the total harmonic distortion and improving power quality. The results are obtained in terms of output voltage waveform, output current waveform, active and reactive output power, SOC of battery system using FFT analysis for THD with and without active filter. The methodology proposed uses simulink on MATLAB environment.

Keywords— active filter, charging station electric vehicle, electric vehicle, harmonics, power quality

I. INTRODUCTION

Maintaining ideal power quality is a burning issue in electric vehicles. This paper examines the inclusion of active filters for power quality improvement at grid level considering EV charging stations. Through the mitigation of disturbances and harmonics, the study tackles the crucial requirement of a strong and effective power infrastructure in the rapidly growing field of electric mobility.

The impending scarcity of fossil fuels and the current environmental constraints of lowering greenhouse gas emissions have prompted substantial research into electric vehicles (EVs) [1]. Consumer desire to move from traditional internal combustion engine vehicles to EVs as an alternative mode of transportation, however, there has been significant impact on EV development. This willingness is the most important component in predicting future EV demand. The authors highlighted that one of the major issues facing the EV market is its charging time [2]. As a result, the focus of this paper is on developing unique ways for minimizing EV charging time by implementing rapid charging rates. In the United States, three levels of EV charging station depicted in “Fig. 1” are being researched and developed. The charging levels of electric vehicles are classed based on their power charging rates [3].

Charging during the night occurs in level I, with the EVs integrated into a handy power outlet (120 V) for sluggish charging (1.5-2.5 kW) over long periods of time. The main disadvantage of level-I charging is the long charging time, which makes it unsuitable for long driving cycles requiring many charging operations. Furthermore, from the perspective of electrical grid operation, the prolonged charging hours at

night overburden distribution transformers, which are not permitted to rest in a grid system with a large number of linked EVs [4]. Level-II charging necessitates a 240 V outlet. As a result, it is frequently utilised as the principle charging method in both private and public facilities. This charging level can restore exhausted EV batteries by giving power in the range of 4 - 6.6 kW over a 3 - 6 hour period. The biggest disadvantage of this charging level is the time required. Furthermore, voltage dips and substantial energy losses in an electrical grid system with a large penetration of level II charging are two of the obstacles that are major limitation. Control and coordination at level II would mitigate the detrimental effects of level-II charge, but this would necessitate the implementation of a comprehensive communication system [5].

In general, single phase power sources with on-board vehicle chargers are required for both levels I and II. Three-phase power systems, on the other hand, are employed with off-board chargers for level III rapid charging rates (50-75 kW). The usage of fast charging stations minimizes the charging time of electric vehicles to less than half an hour for a complete charging cycle. Furthermore, a broad deployment of quick EV charging stations throughout urban and residential areas would address the concern about EV range anxiety. The high power charging rates required for level-III over a short period of time, on the other hand, place a considerable strain on the electrical grid [6].

The current grid infrastructure cannot support the high charging rates required for level-III. As a result, while obtaining the quick charging rates while only depend on the electrical grid, necessitates not only improvements to the charging mechanism, but also increases the demand on the electrical grid [10].

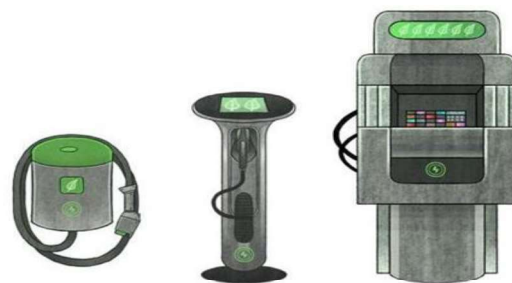


Fig.1. Three levels of chargers for EV charging stations [7].

In EVs solar energy is also used as a source of electricity. This may be in form of solar plate or solar paint [8]. Modern photovoltaic make use of single-crystalline semiconductors manufactured at high temperatures using complex crystal

growth methods on huge wafers with high purity [9]. Organometallic perovskites with millimeter-scale crystalline grains can be grown continuously and without pinholes using a hot-casting approach based on a solution [10-11]. The efficiency of our planar solar cells, which had no cell-to-cell fluctuation, was around 18 percent. Previously, perovskite devices had a hysteresis-inducing photovoltaic response. This has been solved with the new devices. Researchers believe that reduced bulk flaws and improved mobile charging can explain the enhanced performance in large-grain devices [12-13].

The rest of paper is outlined as follows: Section 2 describes various types of charging station, problem formulation is detailed in section 3. While, section 4 contains its solution methodology. MATLAB Simulation and results are discussed in section 5. Finally, the paper is concluded in section 6.

II. TYPES OF CHARGING STATIONS

Electric motors (EVs) have become more popular in India, and to keep them charged, There are special varieties of charging stations. These stations are like gasoline stations for EVs, where it is plugged into the cars or scooter to get it charged up. Here, there are the three main kinds of EV charging stations it might come across in India [14].

A. Slow Charging Stations (AC Charging)

Slow charging stations are like ordinary electricity sockets at domestic. The use of Alternating Current (AC) to charge EV. The slowest kind, this one might fully assess the vehicle in a few hours or even a single day.

People regularly have sluggish chargers at domestic, and only a few workplaces provide them. They are not usually located in public locations. Slow chargers are notable for when one has time, like charging an electric vehicle in a single day at domestic [15].

B. Fast Charging Stations (DC Charging)

Direct Current (DC) is used by fast charging stations to charge EVs. Compared to slow chargers, they are faster. It can typically charge an electric vehicle in a few hours, which is significantly faster than slow chargers.

Fast chargers are frequently found at shopping malls, parking lots, and beside highways, which makes them convenient for quick stops along the way. Fast chargers are ideal for quickly recharging an electric car when you don't have much time to spare or for topping off the battery while being out and about [16].

C. Ultra-Fast Charging Stations (DC Fast Charging)

These charging stations are the fastest of the bunch and additionally use Direct Current like speedy chargers. Ultra-speedy chargers can feed your EV very unexpectedly, frequently within 30 minutes or much less. They are frequently positioned along highways and predominant travel routes for lengthy-distance tour. Ultra-fast chargers are designed for long trips whilst you need a short recharge to continue your experience without ready too lengthy. The form of charging station you operate relies upon to your desires.

- Slow chargers are handy for overnight charging at domestic or work.

- Ultra-speedy chargers are ideal for lengthy-distance journey whilst you want a quick recharge [17].



Fig.2. Different charging stations [19].

The critical power quality issues in integrated grids attributes to EV charging stations are,

1) *Voltage fluctuations* : Voltage fluctuations can come from EV charging stations because of the abrupt need for high power during charging. Voltage sags or swells brought on by abrupt changes in power consumption can affect the quality of power delivered to other connected devices [20].

2) *Current harmonics*: The grid may experience harmonic currents due to the high-power charging apparatus utilized in EV stations. Multiples of the fundamental frequency, or harmonics, have the ability to distort the power supply's sine wave, which may have an impact on other devices and increase distribution system losses [21].

3) *Voltage unbalance*: Voltage imbalance may arise from an uneven load allocation among phases, particularly when several EVs are charging concurrently. This may result in higher losses, lower motor efficiency, and more strain on the power distribution infrastructure [22].

4) *Frequency deviation*: The power grid may experience frequency changes as a result of the abrupt and intense charging activity. Power system equipment must operate at a steady frequency; deviations can cause problems including improper synchronization and sometimes even harm to associated devices [23].

5) *Power factor issues*: EV chargers may have a variable power factor, especially during off-peak hours. Low power factors can increase the reactive power demand, potentially causing inefficiencies in the power distribution system.

6) *Voltage flicker*: Voltage flicker can be brought on by abrupt changes in load, such as those brought on by EV charging. This flickering may damage delicate equipment and be perceived as variations in visible light in lighting systems.[24]

III. PROBLEM FORMULATION

EV chargers generate harmonics, a deteriorated voltage profile, and power loss in distribution transformers since they are non-linear loads. Level II AC charging schemes, with a utmost current rating of 16 A and a utmost power rating of 3.3 kW, are utilized worldwide for EV charging. The majority of electric cars employ single phase 240 V, 50 Hz supply systems and have power gradient between 0.5 and 1 kW [21]. The mathematical modeling for harmonics, voltage level, and

transformer overloading as a result of EV charging in this section.

A. Power Demand

The electric distribution system provides power to the electric vehicle's battery. When nonlinear components are included in the system, the system may become unstable as the demand for power rises. The Eqn. 1 may be used to express an EV's power need.

$$P_{EV} = \frac{C_{Batt.} \times (SOC_{max.} - SOC_{min.})}{T_D} \quad (1)$$

Here, $C_{Batt.}$ is capacity of battery, T_D is time duration of the battery charging. Whether an EV uses low or high power depends on its state of charge (SOC). The total of each EV's individual power demand is the EVs' gross power demand.

$$P_{Gross} = \sum_{N=1}^N P_{EV} \quad (2)$$

B. Harmonics

Harmonics are the increase in voltage and current high frequency components relative to the fundamental frequency. Power quality is impacted by harmonic distortion of the voltage and current waveforms. Total harmonic distortion (THD) of current and voltage is a useful tool for measuring it.

$$THD_i = \sqrt{\frac{\sum_{n=2}^N I_n^2}{I_1^2}} \times 100\% \quad (3)$$

$$THD_V = \sqrt{\frac{\sum_{n=2}^N V_n^2}{V_1^2}} \times 100\% \quad (4)$$

This equation conveys the THD for current and voltage respectively. For slow charging THD_i , THD_V will be less than the fast charging. Thus, the EV with low SOC will have a great chance to produce harmonics.

C. Voltage Profile

When EV charging occurs, the low voltage profile turns into a dangerous problem. When a load suddenly increases or decreases, the power network's capacity to remain steady is referred to as voltage stability. Electric vehicle loads consume a lot of electricity quickly. The grid will become unstable as a result of the degradation of the voltage profile.

D. Transformer Operation

Transformers improve the energy management, battery efficiency, and smart charging capabilities of electric vehicles (EVs). They ensure optimal vehicle functioning by facilitating real-time data processing for predictive maintenance. In the ecosystem of electric transportation, transformer technology in EVs helps to enhance sustainability by enabling faster charging, longer range, and overall better sustainability.

IV. SOLUTION METHODOLOGY USING SIMULATION

The block diagram for the proposed work is depicted in fig. 3. To perform the proposed work mainly components required are AC voltage source, transformer, converters, filter and electric vehicle. The voltage source may be taken from grid supply.

The transformer is used to step down the grid voltage. The work of the converter is to convert grid ac voltage into dc supply for battery charging. The EV is used as load in form of battery system which is going to be charged by charging station.

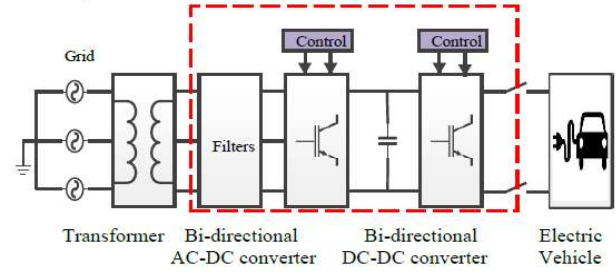


Fig.3. Block diagram of the Proposed system [25].

In the proposed work shunt active filter is used for the mitigation of THD in output of the battery system.

V. RESULTS AND DISCUSSION

The simulation of the proposed is designed in MATLAB Simulink. The main components of this system are ac voltage source, transformer, converters, shunt active filters and battery system used in electric vehicles.

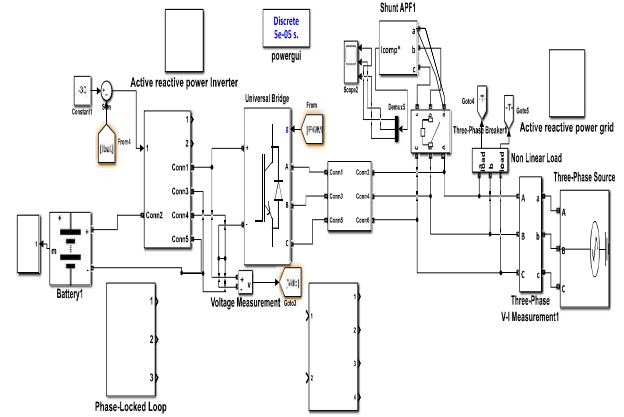


Fig.4. Simulation model of EV charging station.

The results of the proposed work are categorized in two cases. The first case contains the graphs of voltage, current, power, SOC of battery and THD without active filter. The second case contains the graphs of voltage, current, power, SOC of battery and THD with active filter.

Case 1: Charging station without shunt active filter

Fig. 5 showing the graph of voltage and current of charging station before inverter system.

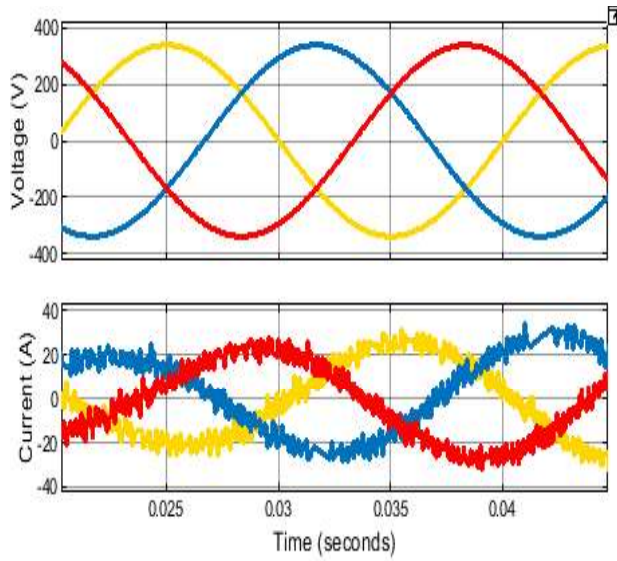


Fig. 5 Source voltage and current.

Fig. 6 showing the graph of active and reactive power of the voltage source.

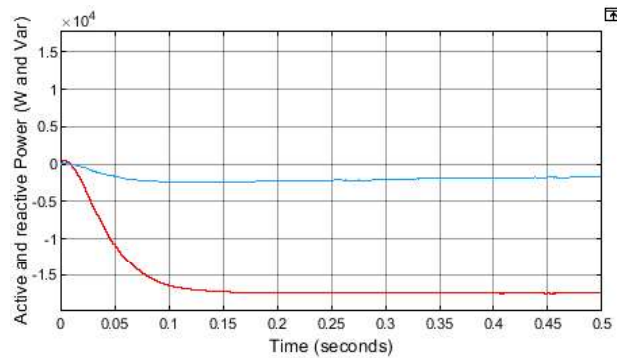


Fig. 6 Active power and reactive power of source.

Fig. 7 showing the SOC graph of battery system. It is showing the charging and discharging of the battery.

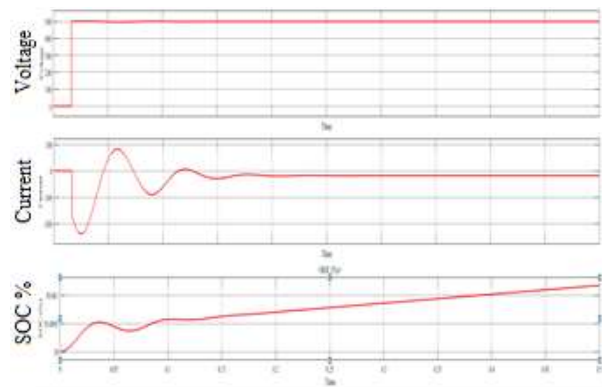


Fig. 7 SOC of battery system.

Fig. 8 showing the THD in output of battery system. It is measured 12.41% without active filter.

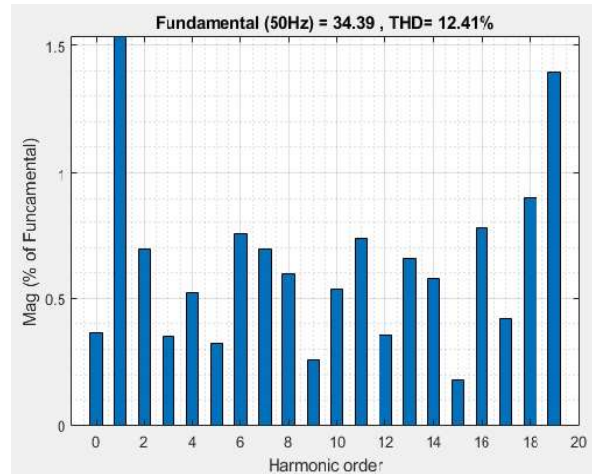


Fig. 8 THD value without active filter.

Case 2: Charging station with shunt active filter

Fig. 9 showing the graph of voltage and current of charging station before inverter system.

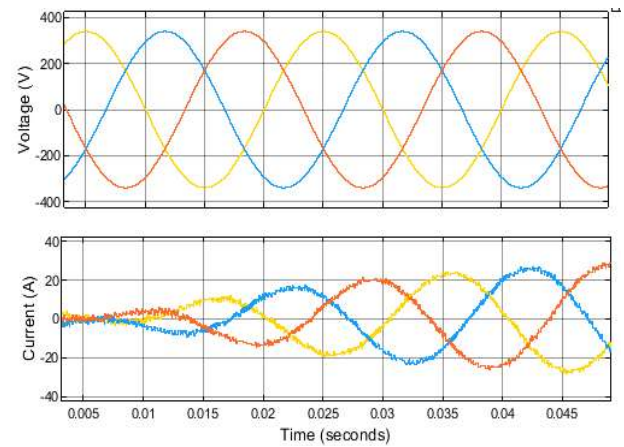


Fig. 9 Voltage and current of power source.

Fig. 10 showing the graph of active power and reactive power of the voltage source.

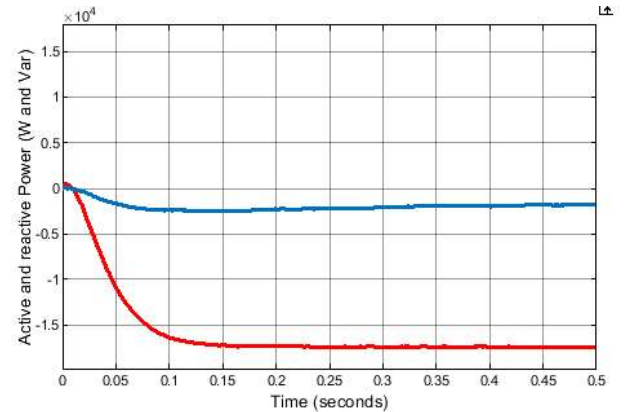


Fig. 10 Active power and reactive power of source.

Fig. 11 showing the SOC graph of battery system. It is showing the charging and discharging of the battery.

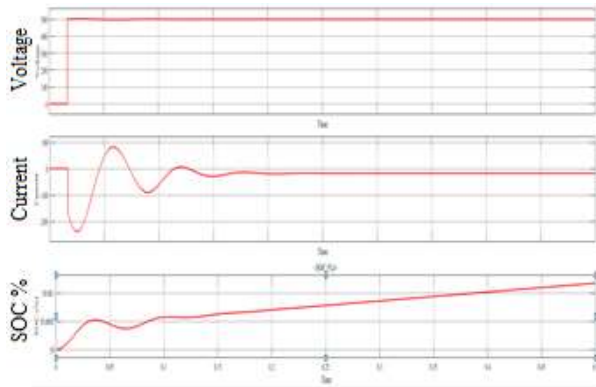


Fig. 11 SOC of battery system.

Fig. 12 showing the THD in output of battery system. It is measured 4.0% with active filter.

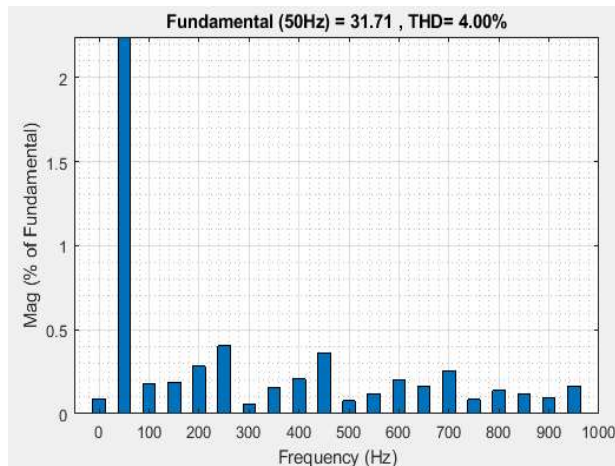


Fig. 12 THD value with active filter.

VI. CONCLUSION

Power quality of the grid is a serious challenge with the advent of electric vehicles. The present paper has dwelled on this issue. In this paper, a Simulink model in MATLAB environment has been developed to address these issues at the charging station level. The proposed model has reduced the THD by using the shunt active filter. The value of the THD has been reduced from 12.41% to 4.0%. Also there has been significant improvement in voltage, active power and reactive power.

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